Class 12 HW: Q13 and Q14

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> Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```r
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

```
   sample geno      exp
1 HG00367  A/G 28.96038
2 NA20768  A/G 20.24449
3 HG00361  A/A 31.32628
4 HG00135  A/A 34.11169
5 NA18870  G/G 18.25141
6 NA11993  A/A 32.89721
```

```r
nrow(expr)
```

```
[1] 462
```

```r
table(expr$geno)
```

```

A/A A/G G/G
108 233 121
```

This result shows the sample size for each genotype.

```
tapply(expr$exp, expr$geno, median)
```

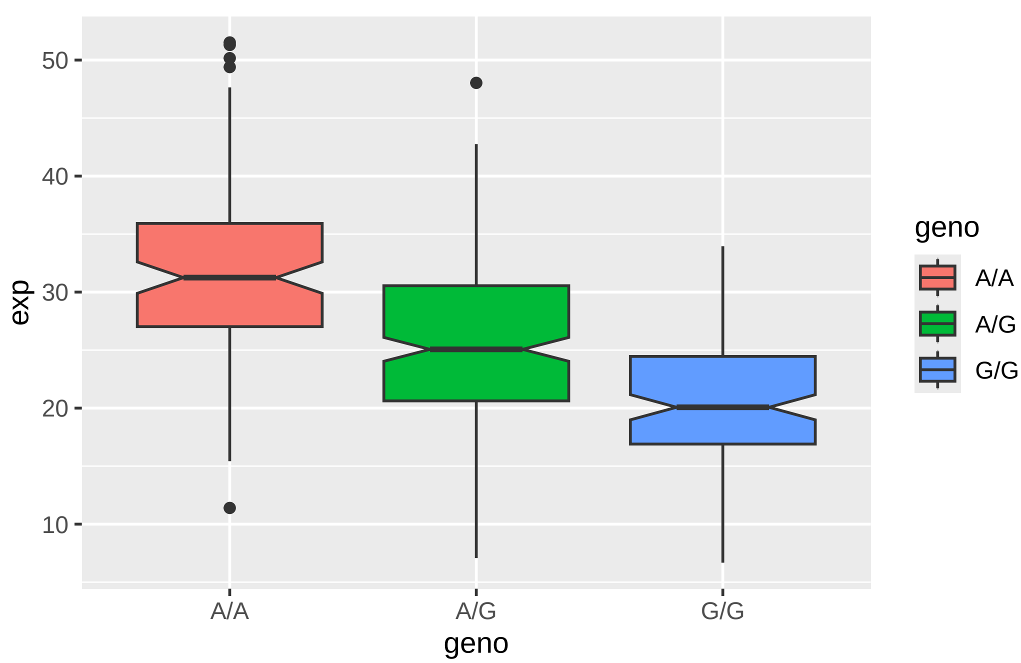
```
     A/A      A/G      G/G
31.24847 25.06486 20.07363
```

This result shows the corresponding median expression level for each genotype.

> Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)
```

```
ggplot(expr) + aes(geno, exp, fill=geno) +
  geom_boxplot(notch=TRUE)
```



From the boxplot, relative expression value between A/A and G/G is clearly different - order of expression level is genotype A/A > A/G > G/G. Thus, the SNP does affect the expression level of ORMDL3.